

HMD-only Controllable 3D Gaussian Avatars in VR: Face and Full-body Demonstration

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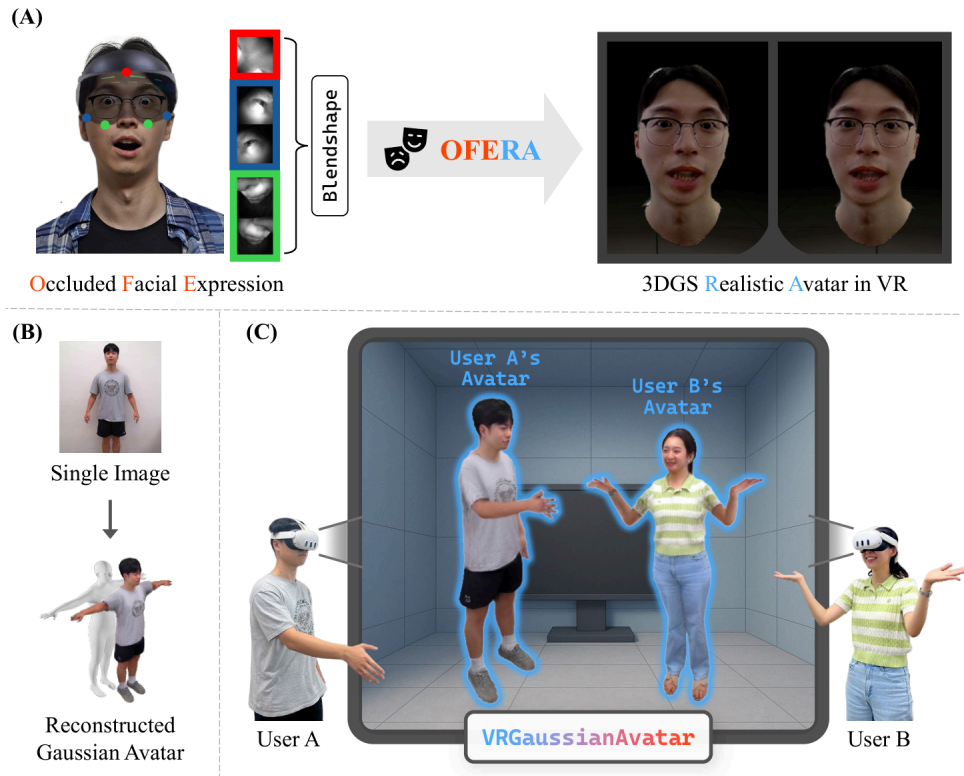


Figure 1: Our two-part VR demonstration system with 3D Gaussian Splatting (3DGS)[2] avatars, consisting of a face part and a full-body part. (A) Face part: HMD-provided blendshape signals drive an expressive 3DGS head avatar under HMD occlusion (OFERA[5]). (B–C) Full-body part: a user-specific full-body Gaussian avatar is reconstructed from a single image and then controlled in VR using HMD-only sensing to support telepresence scenarios (VRGaussianAvatar[3]).

ABSTRACT

We present a VR research demo for practical telepresence with photorealistic 3D Gaussian Splatting (3DGS) avatars under HMD-only sensing. The demo consists of two parts that provide complementary experiences in a single session: a face part, where HMD-provided blendshape signals drive an expressive 3DGS head avatar rendered stereoscopically on the client, and a full-body part, where a user-specific 3DGS full-body avatar is created from a single image and driven using HMD signals. During the demo, participants first capture a single image for full-body avatar creation, experience facial control using a pre-built third-person head avatar while the full-body avatar is generated, and then control their personalized

full-body avatar in VR. By enabling both facial and full-body controllability and contrasting two rendering-responsibility designs—client-side direct 3D rendering and server-side stereoscopic image rendering/streaming—the demo offers practical insights for designing responsive, realistic VR telepresence systems under HMD-only sensing.

Index Terms: Virtual reality, 3D Gaussian avatar, Telepresence

1 INTRODUCTION

Accurately conveying a user's appearance, motion, and facial expressions through an avatar is essential for immersive VR telepresence. A photorealistic avatar that closely resembles the user can strengthen self-identification and social trust in remote interactions, while real-time reproduction of body motion and facial expressions helps communicate intent and affect more clearly. Recent advances in 3D Gaussian Splatting (3DGS)[2] and controllable avatar pipelines that couple deformable neural primitives with explicit control parameters have made it increasingly feasible to render high-quality, responsive avatars in real time, motivating a growing number of VR avatar demos and systems.

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Despite this progress, existing VR avatar demonstrations and telepresence systems often focus on only a single control region—either the face or the body—under practical HMD-only sensing constraints. For example, VOODOO XP[4] primarily targets head and face reenactment for VR telepresence, enabling expressive facial communication but not providing a comparable full-body embodiment experience. On the other hand, body-centric systems such as EgoAvatar[1] emphasize whole-body avatar generation and driving, yet typically do not offer a tightly integrated experience that also includes realistic facial expression control within the same end-to-end demo. As a result, users and system designers have limited opportunities to directly experience both facial and full-body controllability in one coherent VR telepresence workflow, which is important when building practical and user-resembling avatar systems.

In this paper, we present a VR research demo that allows users to experience both facial and full-body avatar control in a single session, while additionally contrasting two practical rendering-responsibility designs for 3DGS avatars in VR. Our demo is composed of two parts. In the face part, we build upon OFERA[5]: facial expressions sensed from an off-the-shelf VR HMD are processed on a server to update the deformation of a 3DGS head avatar, and the updated primitives are transmitted to Unity, where stereoscopic rendering is performed directly for 6DoF viewing. In the full-body part, we build upon VRGaussianAvatar[3]: a user-specific full-body avatar is created from a single image within about one minute, and HMD-only signals are used to drive the avatar while the server renders stereoscopic images that are streamed to the Unity client for display. By structuring the demo into these two complementary experiences, users can compare face versus body controllability as well as client-side direct 3D rendering versus server-side stereoscopic image streaming. This hands-on comparison provides practical insights for designing realistic, responsive VR telepresence systems that rely on HMD-only sensing and neural avatars.

2 SYSTEM DESCRIPTION

Our demo has two parts, Face and Full-body. In both parts, participants observe an avatar that follows their own actions in a mirror-style view. For each part, we summarize the avatar creation input, the HMD-only sensing signals, and the final rendering pipeline. In our implementation, the server and the Unity client run on the same local machine and communicate via localhost under the same computing resources, rather than relying on an external server.

2.1 Face Part

We build the face part on OFERA[5], which produces an expressive 3D Gaussian head avatar. The avatar is created offline from a monocular facial video of a target subject. In our current demo flow, due to the offline construction time, we use a pre-built third-person head avatar rather than generating a participant-specific head avatar on-site. During the demo, a user wears a VR HMD and we capture facial expressions as ARKit-style blendshape coefficients. These signals are sent to a server that maps the blendshapes to a parametric face control space and updates the expression-dependent 3DGS[2] deformation. The updated 3DGS parameters are transmitted to Unity, and Unity renders the head avatar stereoscopically in real time for 6DoF viewing.

2.2 Full-body Part

We build the full-body part on VRGaussianAvatar[3], which produces a 3D Gaussian full-body avatar. A user-specific avatar is created offline from a single image (about one minute in our setup). During the demo, the user wears a VR HMD and we use only device-native signals such as head 6DoF pose and hand tracking. The Unity client estimates a full-body pose from these sparse cues using an inverse-kinematics (IK) module and sends the body pose and stereo camera parameters to a server backend. The backend renders left/right stereoscopic images of the full-body 3DGS avatar, compresses them, and streams them back to Unity for display on the

VR HMD.

Together, the two parts cover both facial and full-body controllability under HMD-only sensing, while illustrating two practical rendering designs: client-side stereoscopic 3DGS rendering for the face part and server-side stereoscopic image rendering/streaming for the full-body part.

3 DEMO WALK-THROUGH

The demo is conducted in three consecutive stages. First, we capture a single image of the participant for the full-body avatar creation. While the full-body avatar is being generated, the participant proceeds to the face part and experiences facial expression control using a pre-built third-person head avatar driven by the same HMD-only sensing pipeline. After the full-body avatar is ready, the participant switches to the full-body part and controls the generated 3DGS[2] full-body avatar using HMD-only signals, observing the result through the stereoscopic rendering/streaming output in VR.

4 CONCLUSION

We present a VR avatar demo designed for practical VR telepresence scenarios. The demo consists of two parts, a full-body avatar experience and a face avatar experience, each implemented with a different rendering-responsibility design. As a result, participants can experience not only both control regions (body and face) but also two distinct approaches to rendering photorealistic avatars in VR. This hands-on comparison provides practical insights into how controllability and rendering design choices affect realistic avatar-based telepresence under HMD-only sensing.

As future work, we plan to provide a unified avatar that supports both full-body motion and facial expression control within a single consistent representation. We also aim to enable live, participant-specific face avatar creation for the demo. In the current system, face avatar construction is time-consuming and thus relies on a pre-built third-person avatar; reducing this creation time would allow generating a personalized 3D head avatar on-site and further improve the telepresence experience.

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